

Image Quality Metrics

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Metrics

- SNR, Dynamic Range
- MTF
- MSE & PSNR, SSIM
- LPIPS, FID
- Delta E

SNR

- SNR is defined to be the ratio of signal to noise.
- Noise adds “uncertainty” to the noise free signal.

Example:

- Your (ideal / perfect) signal generator outputs a pure sinusoid. Your oscilloscope measures a sinusoid with small amplitude variations. The small amplitude variations are attributed to noise.

SNR

- From an image sensor (or photoelectron) perspective:

$$\textit{Model} : y = x + \eta_{shot} + \eta_{read}$$

$$x = E[X] = E[N_{photons} * QE] = \lambda, \text{Var}[X] = \lambda$$

$$QE = \frac{\textit{photoelectrons}}{\textit{incident photons}} = \frac{\textit{photoelectrons}}{N_{photons}}$$

$$\eta_{shot} \sim N(0, \lambda) \text{ and } \eta_{read} \sim N(0, \sigma_{read}^2)$$

$$\rightarrow SNR_{\textit{sensor or photoelectron}} = \frac{\textit{signal}}{\textit{noise}} = \frac{\lambda}{\sqrt{\lambda + \sigma_{read}^2}}$$

SNR

- The noise / noise model is a function of illumination.
- For very low light, read noise dominates: $SNR_{sensor} \sim \frac{\lambda}{\sigma_{read}}$
- Typically, read noise is 2-3e, where e denotes photoelectrons.
- Otherwise, shot noise dominates: $SNR_{sensor} \sim \frac{\lambda}{\sqrt{\lambda}}$
- Shot noise increases with increasing illumination.
- Since SNR increases with increasing illumination, shot noise is only problematic in low light scenarios.

Read Noise

- Read noise is created from the analog electronics, PRIOR to ADC conversion.
- After the photoelectrons are created, the associated charge is converted to a voltage.
- A source follower (then) amplifies this voltage.
- A column amplifier (then) amplifies the voltage from the source follower.

Read Noise

- The column amplified voltage is then measured by the ADC.
- Note: The ADC introduces an additional noise term - quantization noise. Quantization noise depends on the ADC quantization step size (LSB).
- By construction, the units for read noise is volts.
- To convert read noise voltage to photoelectrons, the read noise voltage needs to be divided by a conversion factor c .
- C has units V/e^- .

Read Noise

- In practice, if you are read noise limited, the noise distribution is a skewed gaussian.

Why?

- Gaussian distributions have long tails.
- But, photoelectron counts cannot go negative.
- Hence, the distribution clips.

SNR

- Q: Does analog gain (=voltage gain) affect the SNR?
- A: It depends....

photons $\rightarrow$ photoelectrons $\rightarrow$ analog gain, $g_a c$

$$y = g_a c x + g_a c \eta_{shot} + g_a c \eta_{read}$$

c is a conversion gain with units $\frac{V}{e^-}$

$$SNR_{analog} = \frac{g_a c \lambda}{\sqrt{(g_a c)^2 \lambda + (g_a c)^2 \sigma_{read}^2}} = \frac{\lambda}{\sqrt{\lambda + \sigma_{read}^2}} \rightarrow NO$$

SNR

- Q: Does analog gain (=voltage gain) affect the SNR?
- A: It depends....

photons $\rightarrow$ photoelectrons $\rightarrow$ analog gain, $g_a c \rightarrow ADC \rightarrow ISP$

$$y = g_a c x + g_a c \eta_{shot} + g_a c \eta_{read} + \eta_{quant} \quad \text{and} \quad \eta_{quant} \sim Unif(-\Delta_v/2, +\Delta_v/2), \quad \sigma_{quant}^2 = \frac{\Delta_v^2}{12}$$

$$\rightarrow SNR_{ADC} = \frac{g_a c \lambda}{\sqrt{(g_a c)^2 \lambda + (g_a c)^2 \sigma_{read}^2 + \sigma_{quant}^2}} \rightarrow YES$$

SNR

- Different SNR's exist at different locations:

$SNR_{\text{sensor or photoelectrons}}$, SNR_{analog} , SNR_{ADC}

- Although analog gain can appear to visually improve SNR, it does so, by negating the degradation associated with quantization noise.
- In practice, the only way to truly improve SNR, is by creating more photoelectrons.

SNR

This is accomplished with:

- 1) Longer exposure times
- 2) Larger pixel sizes
- 3) ***Larger apertures, f#
- 4) Brighter scenes
- 5) Higher quantum efficiency

SNR

- What about digital gain?

photons $\rightarrow$ photoelectrons $\rightarrow$ analog gain, $g_a c \rightarrow$ ADC $\rightarrow$ digital gain, $g_d \rightarrow$ ISP

$$y = g_a c g_d x + g_a c g_d \eta_{shot} + g_a c g_d \eta_{read} + g_d \eta_{quant}$$

$$\rightarrow SNR_{digital} = \frac{g_a c g_d \lambda}{\sqrt{(g_a c g_d)^2 \lambda + (g_a c g_d)^2 \sigma_{read}^2 + g_d^2 \sigma_{quant}^2}} \rightarrow NO$$

SNR

- Q: Does increasing the camera ISO (=camera gain) increase SNR?
- A: No (Sensor) and Yes (Analog) and Partially (Analog and Digital Gain)
- A: No (Sensor) You have only increased the amplification = gain. The image became brighter. You did not create more photoelectrons.
- A: Yes (Analog) You have made the quantization noise from the ADC negligible. i.e. You have raised the voltage (attributed to the photoelectron count) so that it exceeds the minimal quantization step = LSB, producing a visual improvement.
- A: Partially (Analog and Digital Gain) Changing the ISO from 100 to 400 typically increases the analog gain. However, changing the ISO from 800 to 3200 increases the digital gain - since the analog gain is already “maxed out”.

Dynamic Range

$$\text{Sensor Dynamic Range}(dB) = 20\log_{10} \left(\frac{\text{Full Well Capacity}}{\text{Read Noise} = \text{Noise Floor}} \right)$$

$$\text{Sensor Dynamic Range}(stops) = 20\log_2 \left(\frac{\text{Full Well Capacity}}{\text{Read Noise} = \text{Noise Floor}} \right)$$

- Full Well Capacity measure the maximum number of photoelectrons that can be stored at a pixel location.
- Full Well Capacity is determined by pixel size.
- In the above formulas, all units are in photoelectrons.

SNR & Dynamic Range

- Q: How does ISO affect the sensor dynamic range?
- A: A higher ISO setting results in a larger gain. For a given number of photoelectrons, the higher ISO will scale the photoelectron count, thereby reducing the dynamic range.

Example:

- ISO 100, Gain = 1, 10,000 photoelectrons
- ISO 400, Gain = 4, 40,000 effective photoelectrons

***f#

$$f\# = \frac{\text{Focal Length}}{\text{Diameter}}$$

- Cell phone cameras have significantly faster lenses than their SLR counterparts.
- Q: Why is this so? Isn't the "speed" of the lens, strongly correlated with the price?
- A:

Cell phone lens f# = f1.7, (50\$?), Professional SLR lens f# = f2.0 (2000\$)

Cell phone focal length = ~5mm, cell phone aperture = ~3mm

SLR focal length = 35mm, SLR aperture = ~17mm

***f#

- In practice, cell phone camera lenses can be more efficient than their SLR counterparts = more light captured per mm².
- If true, cell phone camera lenses will have have a better f# spec.
- However, because of their significantly larger apertures, SLR camera lenses will have vastly superior light gathering capabilities.
- In practice, it is very challenging (= \$\$\$) to make large aperture lenses with premium lens specifications.

The Merits of a Large Sensor

- Large sensors permit larger pixels.
- Larger pixels =

Better dynamic range, because the larger pixels have larger full well capacities.

Better low light performance, because the larger pixels have a larger surface area to collect the incoming photons.

The Downside of a Large Sensor

- For a fixed sensor size, larger pixels will result in a lower image resolution.

=> Best of all worlds = larger pixels and a larger sensor to scale. This will allow you to maintain, if not increase, the image resolution.

=> If size, power, and cost are “non issues”, make everything BIG.

SNR and Dynamic Range (Further Down the Imaging Stack)

- Single frame SNR vs multi frame SNR
- Single frame dynamic range vs multi frame dynamic range

SNR and Dynamic Range (Further Down the Imaging Stack)

Multi frame SNR from ISP

- Consider the simplest case - strict averaging of IID signals using N consecutive frames.
- Let the signal be S.
- Now, the noise decreases by $\frac{\sigma^2}{N}$
- $SNR_{single} = \frac{S}{\sigma}, \quad SNR_{multiple} = \frac{S}{\frac{\sigma}{\sqrt{N}}} = SNR_{single} \sqrt{N}$

SNR and Dynamic Range (Further Down the Imaging Stack)

Multi frame HDR from ISP

- Given the dynamic range of a single frame: DR_{single}
- Given the shortest and longest exposure times from subsequent frames:
 t_{long} and t_{short} , $R = \frac{t_{long}}{t_{short}}$

$$DR_{new} = DR_{single} R \rightarrow 20\log(DR_{new}) = 20\log(DR_{single}) + 20\log(R)$$

- You get approximately a 6dB improvement in DR per stop.

SNR and Dynamic Range (Further Down the Imaging Stack)

- Combining multi frames does not come for free.

Given local motion in the scene:

- Temporal denoising will produce blur.
- Multi frame HDR will produce ghosts.
- Hence, additional algorithms need to be added, to mitigate these effects.

Modulation Transfer Function (MTF)

- From an optics standpoint, we measure the MTF of the lens:
 MTF_{lens}
- This requires individuals with specialized training and access to specialized lab equipment = an optical bench.
- In practice, the lens is one component, in a larger system.
- For example, pixel information is captured by an image sensor behind the lens.
- Then, the pixel information is processed by algorithms in the ISP.

MTF

- Hence, we have: $MTF_{system} = MTF_{lens} MTF_{diffraction} MTF_{pixel} MTF_{sampling} MTF_{ISP}$
- If the output image is extracted in RAW (before MTF_{ISP}) , the MTF is guaranteed to be ≤ 1 .
- However, if the output image is extracted in JPEG (after MTF_{ISP}), the MTF can be > 1 .
- This is because ISP processing - such as sharpening, can amplify the MTF.

MTF

- From a lens-sensor perspective, contrast is defined as:

$$\% \text{ Contrast} = \left[\frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} \right]$$

- By construction, contrast can only degrade. i.e. $\text{MTF} \leq 1$.

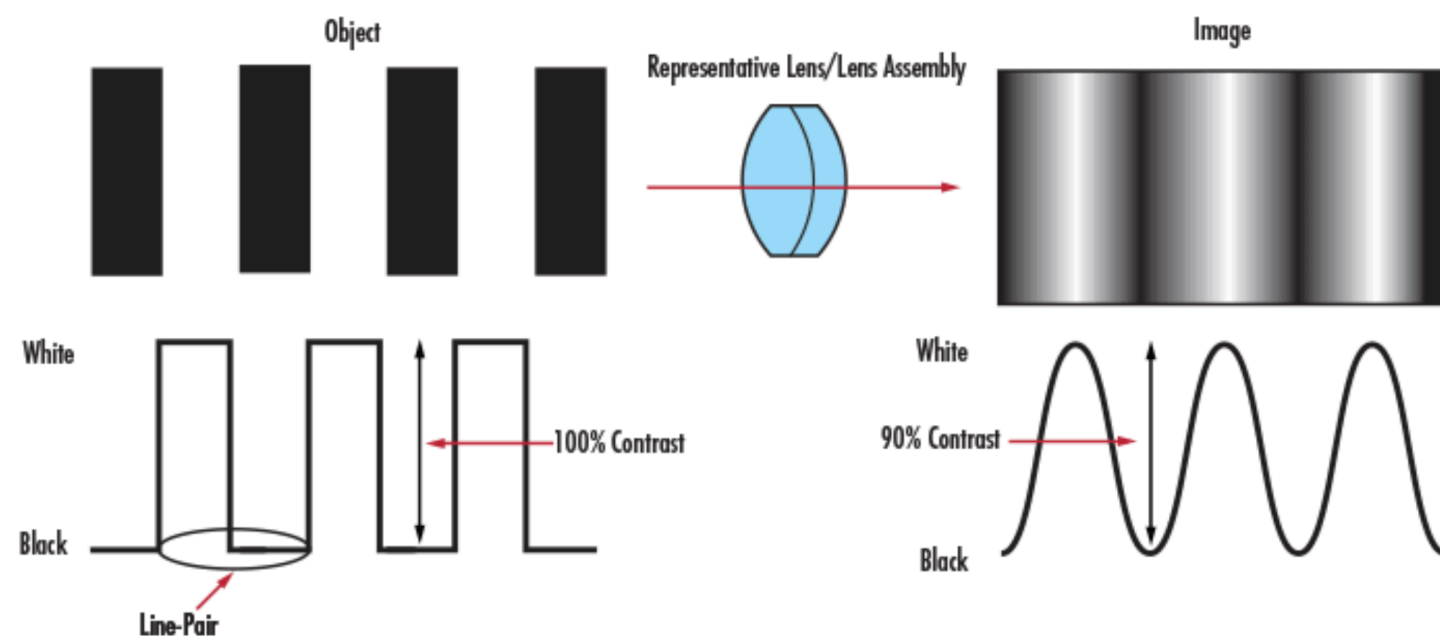
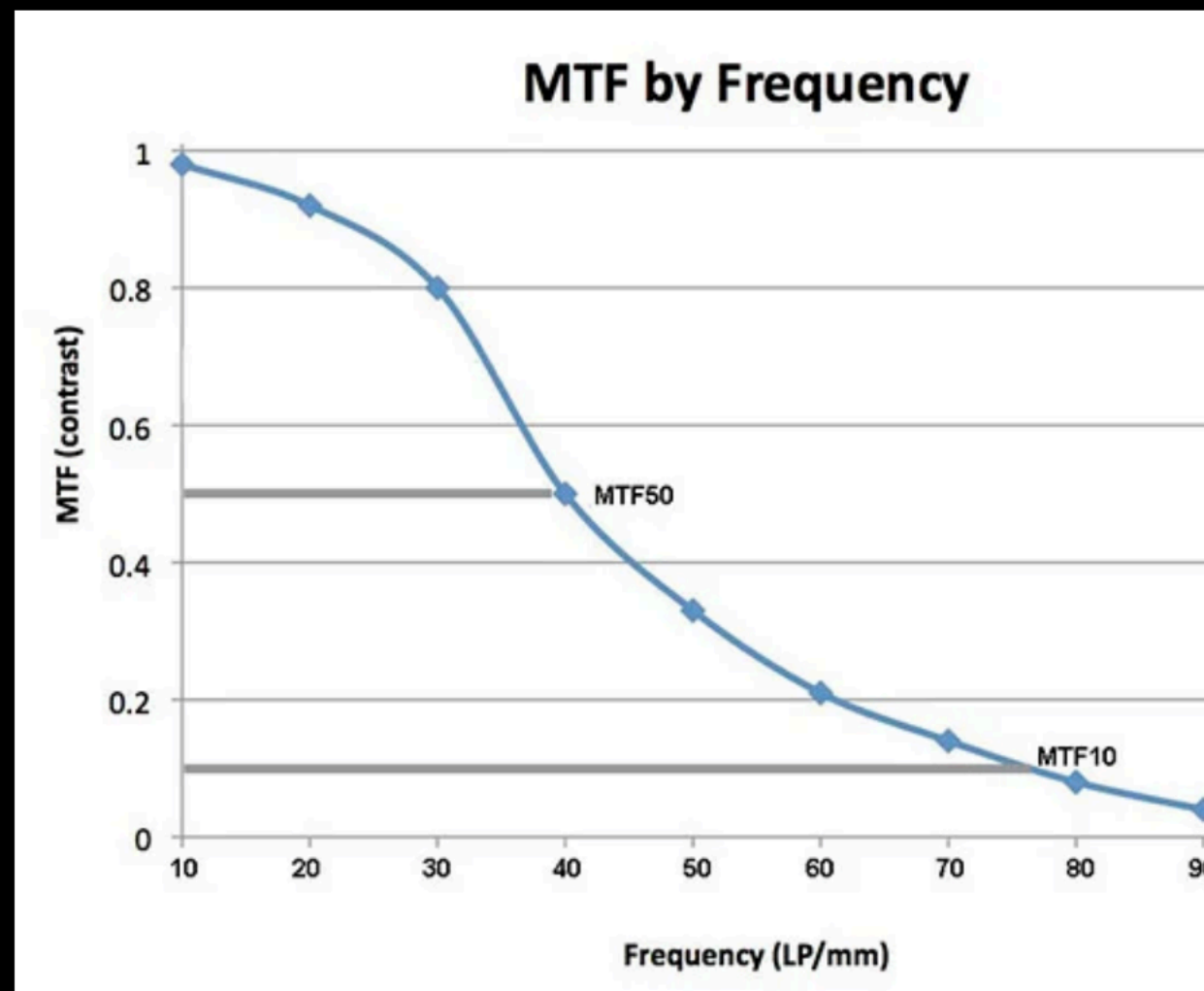


Figure 4: Contrast of a Bar Target and Its Image

MTF

- The MTF plot compares Contrast vs (cycles / mm, line pairs / mm)

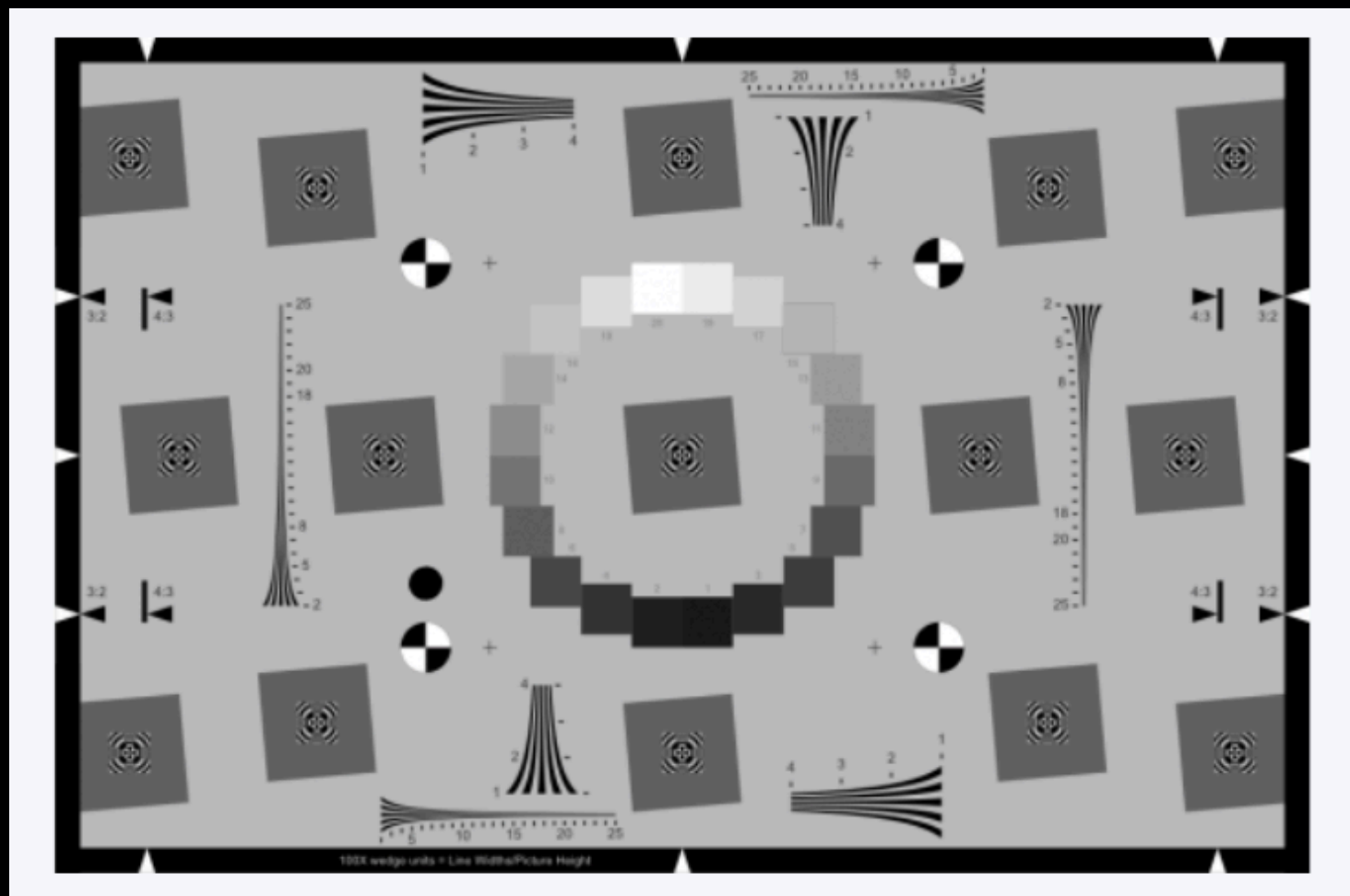


MTF

- Qualitatively - As the line pairs become closer and closer, the contrast degrades.
- MTF50 denotes the the line pair / mm where the contrast has degraded by 50%.
- MTF30 denotes the line pairs / mm where the contrast has degraded to 30%.
- In practice, engineers uses test software (like Imatest) to generate MTF vs line pair / mm plots.
- These plots are generated in the following manner:

MTF

A slanted edge test chart is captured.



MTF

- The software extracts the Edge Spread Function (ESF) from the square(s) in the test chart.
- Differentiation is performed on the Edge Spread Function to obtain the Line Spread Function (LSF).
- The FFT of the LSF is computed, yielding the OTF.
- The magnitude of the the OTF is computed, yielding the MTF curve.

MTF

Notes

- The MTF curve is sensitive to edge orientation.
- Hence, the MTF curve will be different for a slant edge at 5 degrees versus an MTF curve with a slant edge at 30 degrees.
- In a perfect world / with a perfect lens, only a single MTF curve would be needed = orientation independent.

MTF

Other comments

- Lens manufacturers (Canon, Nikon, Sony, etc.) release MTF curves for their lenses.
- These MTF curves are different from the MTF curve just described.
- Specifically, lens manufacturers measure how their lens degrade spatially, from the center of the lens.
- We mention the lens manufacturer MTF curves, for the sake of completeness.

MTF

- From an engineering standpoint, we can obtain a reasonable approximation to the lens MTF via:

$$MTF_{lens} = \frac{MTF_{system}}{MTF_{diffraction} MTF_{pixel} MTF_{sampling}}$$

To simplify the future analysis, let's assume:

1) Square pixels

2) Sensor fill factor = 100% => pixel pitch = pixel width = p

- Of great importance, is the Nyquist frequency.
- This is determined by the pixel sampling rate and is given by: $\frac{1}{2p}$

MTF

- MTF cannot be measured beyond Nyquist. i.e. $f < f_{Nyquist}$

$$f_{Nyquist} = \frac{1}{2p}$$

$$f_{diffraction\ cutoff} = \frac{1}{\lambda N}, \quad N \text{ is the } f \text{ number of the lens}$$

$$MTF_{pixel}(f) = |sinc(\pi fp)|$$

- As we stop down the lens - resulting in a larger f number, the lens PSF increases due to increased diffraction.
- In the frequency domain, the diffraction cutoff then decreases.
- At the Nyquist frequency, $MTF_{pixel}(f_{Nyquist}) = |sinc(\pi fp)| = |sinc(\frac{\pi}{2})| \approx 0.64$

MTF

- From a lens MTF measurement standpoint, you want the contributions from $f_{Nyquist}$ and f_{pixel} to be small relative to $f_{diffraction}$
- This requires that the pixel pitch and the pixel size be small.
- From a system standpoint, this combination significantly reduces the SNR.
- From a system design standpoint, you want $f_{Nyquist} \approx f_{diffraction\ cutoff}$

MTF

Focusing solely on the lens:

- Large aperture lenses experience little degradation due to diffraction.
- Hence, the lens MTF is dominated by lens aberrations.
- Small aperture lenses become diffraction limited.
- Lens aberrations are now insignificant, at the expense of a significantly reduced lens diffraction cutoff frequency.

MTF

- For mobile phone cameras, $f_{Nyquist} < f_{diffraction\ cutoff}$
- However, the influence of diffraction increases for mobile phone tele lenses, due to their larger f#.

MSE & PSNR

- MSE and PSNR are two commonly used metrics in imaging.

$$MSE = \frac{1}{MN} \sum_{i=1}^M \sum_{j=1}^N \left(I(i, j) - \hat{I}(i, j) \right)^2, \quad PSNR = 20 \log_{10} \frac{Max_I}{\sqrt{MSE}}$$

- Because of their known shortcomings, advanced metrics like SSIM were proposed.

SSIM

- SSIM is a perceptual metric, used to evaluate image quality.
- Prior to SSIM, metrics focused on quantifying the “visibility of errors” between a reference image and a distorted image.
- For example, mean square error (MSE) computed the per pixel squared error.



(a)



(b)



(c)



(d)



(e)



(f)

Fig. 2. Comparison of “Boat” images with different types of distortions, all with $MSE = 210$. (a) Original image (8bits/pixel; cropped from 512×512 to 256×256 for visibility); (b) Contrast stretched image, $MSSIM = 0.9168$; (c) Mean-shifted image, $MSSIM = 0.9900$; (d) JPEG compressed image, $MSSIM = 0.6949$; (e) Blurred image, $MSSIM = 0.7052$; (f) Salt-pepper impulsive noise contaminated image, $MSSIM = 0.7748$.

b), c), d), e), f) all have the same MSE, yet some errors are more objectionable than others.

SSIM

- The authors of SSIM pointed out that existing metrics were “visibility based”, bottoms up approaches.
- The authors argued that a top down approach, based on image structure, would be a valuable way to evaluate image quality.
- As a result, the authors developed the structural similarity metric (SSIM).

SSIM

$$SSIM(\overrightarrow{x}, \overrightarrow{y}) = [l(\overrightarrow{x}, \overrightarrow{y})]^\alpha \cdot [c(\overrightarrow{x}, \overrightarrow{y})]^\beta \cdot [s(\overrightarrow{x}, \overrightarrow{y})]^\gamma$$

The 3 terms measure differences between the distorted image and the reference image.

- The first term measures the difference in luminance.
- The second term measures the difference in contrast.
- The third term measures the degree of correlation.

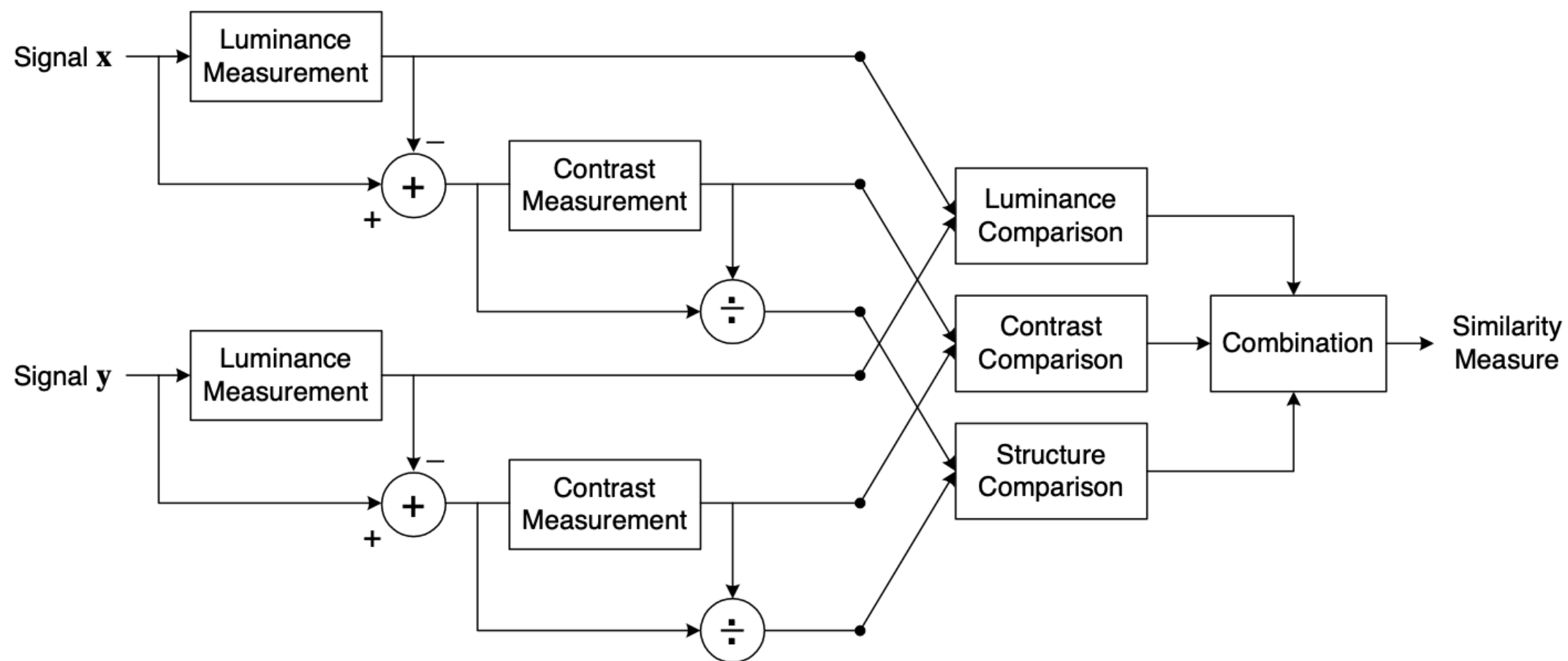


Fig. 3. Diagram of the structural similarity (SSIM) measurement system.

SSIM

On a local patch:

- The difference in luminance is given by:

$$l(\overrightarrow{x}, \overrightarrow{y}) = \frac{2u_x u_y + C_1}{u_x^2 + u_y^2 + C_1}$$

- The difference in contrast is given by:

$$c(\overrightarrow{x}, \overrightarrow{y}) = \frac{2\sigma_x \sigma_y + C_2}{\sigma_x^2 + \sigma_y^2 + C_2}$$

- The degree of correlation is given by:

$$s(\overrightarrow{x}, \overrightarrow{y}) = \frac{\sigma_{xy} + C_3}{\sigma_x \sigma_y + C_3}$$

$C_1, C_2, C_3, \alpha, \beta, \gamma$ are user defined constants .

SSIM

- SSIM is computed by averaging the results from M spatial patches.
- The metric was applied to JPEG compressed images, illustrating its efficacy.
- More recently, the SSIM metric was challenged in: Understanding SSIM, Nilsson and Akenine-Moller, arXiv 2020.
- Specifically, the authors question the efficacy of using SSIM to train deep networks.

Frechet Inception Distance (FID)

- FID is used to measure the closeness / similarity in distributions.
- The FID formula / score results, after one distribution has undergone an “optimal transport” mapping to the other distribution.
- FID finds widespread use in image generation, to determine the closeness of a reference image test set to a generated image test set.

Frechet Inception Distance (FID)

Overview

- For every input image in a test set, compute a 2048D output feature using an ImageNet trained inception network.
- Compute the mean feature vector from the output feature vectors.
- For the every input image in the same test set, compute the covariance matrix of the 2048D feature vector.
- Compute the mean covariance matrix (2048Dx2048D) from the output covariance matrices.

Frechet Inception Distance (FID)

- Do this, for (both) the reference test set and the generated test set.
- What you have computed:

$\vec{u}_r, \vec{u}_g$ are the mean feature vectors with dimension $2048D$

Σ_r, Σ_g are the mean covariance matrices with dimension $2048D \times 2048D$

$$FID \triangleq ||u_r - u_g||^2 + Tr \left(\Sigma_r + \Sigma_g - 2 \left(\Sigma_r \Sigma_g \right)^{\frac{1}{2}} \right)$$

- A low FID value indicates a good match between the reference and generated test set feature distributions after “optimal transport”.

Frechet Inception Distance (FID)

High Level physical picture

- The reference and generated distributions can be viewed as multidimensional ellipses.
- The FID score assigns a number to the best match that can be obtained between the two ellipses, after an “optimal transport” is applied to one of the ellipses.
- If the ellipse centroids match, the ellipse shapes match, and the ellipse shapes are aligned (via rotation), the FID score is 0.

Frechet Inception Distance (FID)

In practice:

- We assume that we have two multivariate gaussian distributions - one for the reference test set and one for the generated test set.
- We want to measure the difference between the two distributions, after an optimal transport mapping.
- For gaussian distributions, this “optimal transport” is given by an affine matrix A .

1) In practice, the multivariate gaussian distributions can / will differ in their centroid locations.

2) In practice, the multivariate gaussian distributions can / will differ in their orientation and shape.

Frechet Inception Distance (FID)

- The FID equation results from: $\text{minimize } E[||X - T(X)||^2]$
- Let X represent the reference distribution.
- We apply transformation / transport T that maps X closest to the generated distribution.
- Since the reference and the generated distributions are gaussian, the optimal transport T is given by: $T(x) = u_g + A(x - u_r)$
- ... where A is an affine matrix.

Frechet Inception Distance (FID)

$$T(x) = u_g + A(x - u_r)$$

$$\text{For } X \sim N(u_r, \Sigma_r) \rightarrow X - u_r \sim N(0, \Sigma_r)$$

T(X) has mean u_g

$$\text{Cov}(T(X)) = A \Sigma_r A^T$$

$$\text{We want } T(X) \sim N(u_g, \Sigma_g)$$

$$\text{This occurs when } A \Sigma_r A^T = \Sigma_g$$

Frechet Inception Distance (FID)

- Key: Find the optimal affine matrix A that satisfies this requirement.
- Recall: $\text{minimize } E[||X - T(X)||^2] \rightarrow E[||u_r - u_g + (I - A)(X - u_r)||^2]$
- After a lot of math (omitted to improve clarity), the optimal affine matrix A produces the FID equation:

$$||u_r - u_g||^2 + \text{Tr}\left(\Sigma_r\right) + \text{Tr}\left(\Sigma_g\right) - 2\text{Tr}\left(\Sigma_r^{\frac{1}{2}}\Sigma_g\Sigma_r^{\frac{1}{2}}\right)^{\frac{1}{2}}$$

Frechet Inception Distance (FID)

What the equation says:

- 1) There is a penalty when the centroids of the two distributions do not match.
- 2) The reference and generated distributions have a size and shape, given by the trace terms of their respective covariance matrices.
- 3) If the eigenvalues and eigenvectors of the reference and generated distribution are the same, the third trace term will cancel the first and second trace terms.
- 4) Otherwise, a difference will exist, despite the best match (affine transform) provided by A .

ΔE

- ΔE is a perceptual color metric, measuring perceptual color difference.

- i.e. How different do two colors look to a human?

$\Delta E = 0$, *Identical*

$\Delta E = 1$, *Barely Perceptible*

$\Delta E > 3$, *Noticeable*

- ΔE is computed in a perceptually uniform color space - Lab.

- Original - $\Delta E_{76} = \sqrt{(L - L_{ref})^2 + (a - a_{ref})^2 + (b - b_{ref})^2} = \sqrt{\Delta L^2 + \Delta a^2 + \Delta b^2}$

- Modern - $\Delta E_{00} = \sqrt{\left(\frac{\Delta L'}{S_L}\right)^2 + \left(\frac{\Delta C'}{S_C}\right)^2 + \left(\frac{\Delta H'}{S_H}\right)^2 + R_T \left(\frac{\Delta C'}{S_C}\right) \left(\frac{\Delta H'}{S_H}\right)}$

$$\Delta E$$

S_L, S_C, S_H are weighting terms.

ΔE_{00} adds adaptive scaling and correction terms across lightness, chroma, and hue, relative to ΔE_{76}

In general:

- The human visual system is non-linear.
- The human visual system sensitivity differs across hues.
- i.e. The human visual system is more sensitive to green shifts than blue shifts.

$$\Delta E$$

- Hence, computing a Euclidean distance across R, G, and B (as an error metric) produces a biased result:

$$\sqrt{(R - R_{ref})^2 + (G - G_{ref})^2 + (B - B_{ref})^2}$$

- Note: The reference values are illuminant dependent.
- Hence, a ΔE vs *Illuminant* plot is needed to obtain the complete picture.



Data

Original Munsell Notation CIE 1976 L*a*b*, D50, 2°					
Name	Hue	Value/Chroma	L*	a*	b*
Dark skin	3.05 YR	3.69/3.20	39.07	13.70	14.37
Light skin	2.2 YR	6.47/4.10	66.36	18.08	18.56
Blue sky	4.3 PB	4.95/5.55	50.66	-4.84	-21.44
Foliage	6.65 GY	4.19/4.15	44.03	-13.01	22.34
Blue flower	9.65 PB	5.47/6.70	55.87	8.88	-24.79
Bluish green	2.5 BG	7/6	71.47	-32.98	0.65
Orange	5 YR	6/11	62.37	36.05	56.58
Purplish blue	7.5 PB	4/10.7	40.77	9.18	-43.58
Moderate red	2.5 R	5/10	51.68	48.21	16.75
Purple	5 P	3/7	31.27	20.00	-20.83
Yellow green	5 GY	7.08/9.1	72.74	-22.60	57.15
Orange yellow	10 YR	7/10.5	72.35	19.57	68.76
Blue	7.5 PB	2.90/12.75	29.61	13.63	-49.65
Green	0.1 G	5.38/9.65	55.61	-37.19	31.81
Red	5 R	4/12	42.23	55.32	27.45
Yellow	5 Y	8/11.1	82.65	4.44	80.52
Magenta	2.5 RP	5/12	52.55	49.33	-14.42
Cyan	5 B	5/8	52.24	-28.24	-27.14
White	N	9.5/	96.37	-0.30	3.26
Neutral 8	N	8/	81.70	-0.56	0.25
Neutral 6.5	N	6.5/	66.49	-0.33	0.03
Neutral 5	N	5/	50.67	-1.06	-0.19
Neutral 3.5	N	3.5/	36.23	-0.48	-0.26
Black	N	2/	20.68	0.17	-0.55

$$\Delta E$$

ΔE_{avg} measures the average result from all 24 Macbeth color checker patches

ΔE_{max} is the worst ΔE result amongst all 24 patches

ΔE_{skin} measures the average result from patches 1 and 2.

The deep blue patch often produces the worst result, because the blue spectral sensitivity curve

- 1) Is typically broad (with large overlap into green)
- 2) Does not closely match the human visual system cone response
- 3) Has low quantum efficiency

Summary

- We have presented different metrics of significant importance to imaging / vision.
- SNR and Dynamic Range are sensor related metrics, used to evaluate the purity and range of the image signal.
- MTF is a lens and system metric, used to evaluate the sharpness of captured images.
- FID, SSIM are objective metrics, used to evaluate the efficacy of imaging and vision algorithms.
- ΔE is a perceptual metric, specifically designed to evaluate image color.